MEDIUM · ADVANCED MATH · ANSWER KEY

Advanced Math

11

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1

B

B. 2

Choice B is correct. Since $\frac{x+5}{x+5}$ is equivalent to 1, the right-hand side of the given equation can be rewritten as $\frac{x+5}{x+5}-\frac{1}{x+5}$, or $\frac{x+4}{x+5}$. Since the left- and right-hand sides of the equation $\frac{2(x+1)}{x+5}=\frac{x+4}{x+5}$ have the same denominator, it follows that $2(x+1)=x+4$. Applying the distributive property of multiplication to the expression $2(x+1)$ yields $2(x)+2(1)$, or $2x+2$. Therefore, $2x+2=x+4$. Subtracting x and 2 from both sides of this equation yields $x=2$.

Choices A, C, and D are incorrect. If $x=0$, then $\frac{2(0+1)}{0+5}=1-\frac{1}{0+5}$. This can be rewritten as $\frac{2}{5}=\frac{4}{5}$, which is a false statement. Therefore, 0 isn't a solution to the given equation. Substituting 3 and 5 into the given equation yields similarly false statements.

Q2

B

B. $(x^2+2)(x^2-3)$

Choice B is correct. The term x^4 can be factored as $(x^2)(x^2)$. Factoring –6 as $(2)(-3)$ yields values that add to –1, the coefficient of x^2 in the expression.

Choices A, C, and D are incorrect and may result from finding factors of –6 that don't add to the coefficient of x^2 in the original expression.

Q3 B

B. -57

Choice B is correct. For a quadratic function defined by an equation of the form $px = ax - h^2 + k$, where a, h, and k are constants and $a > 0$, the minimum value of the function is k. Subtracting 57 from both sides of the given equation yields $px = x^2 - 57$. This function is in the form $px = ax - h^2 + k$, where $a = 1$, $h = 0$, and $k = -57$. Therefore, the minimum value of the function p is -57.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4 C

C. Two

Choice C is correct. It is given that the constants a and c are both positive; therefore, the graph of the given quadratic equation is a parabola that opens up with a vertex on the y-axis at a point below the x-axis. The graph of the second equation provided is a horizontal line that lies above the x-axis. A horizontal line above the x-axis will intersect a parabola that opens up and has a vertex below the x-axis in exactly two points.

Choices A, B, and D are incorrect and are the result of not understanding the relationships of the graphs of the two equations given. Choice A is incorrect because the two graphs intersect. Choice B is incorrect because in order for there to be only one intersection point, the horizontal line would have to intersect the parabola at the vertex, but the vertex is below the x-axis and the line is above the x-axis. Choice D is incorrect because a line cannot intersect a parabola in more than two points.

Q5

4

The correct answer is 4. It’s given that $2x^2+5x-12$ can be rewritten as $(2x-3)(x+k)$; it follows that $(2x-3)(x+k)=2x^2+5x-12$. Expanding the left-hand side of this equation yields $2x^2+2kx-3x-3k=2x^2+5x-12$. Subtracting $2x^2$ from both sides of this equation yields $2kx-3x-3k=5x-12$. Using properties of equality, $2kx-3x=5x$ and $-3k=-12$. Either equation can be solved for k. Dividing both sides of $-3k=-12$ by -3 yields $k=4$. The equation $2kx-3x=5x$ can be rewritten as $x(2k-3)=5x$. It follows that $2k-3=5$. Solving this equation for k also yields $k=4$. Therefore, the value of k is 4.

Q6

A

A. The length of the rectangle, in feet

Choice A is correct. It's given that a rectangle has a length that is 15 times its width. It's also given that the function $y = 15ww$ represents this situation, where y is the area, in square feet, of the rectangle and $y > 0$. The area of a rectangle can be calculated by multiplying the rectangle's length by its width. Since the rectangle has a length that is 15 times its width, it follows that w represents the width of the rectangle, in feet, and $15w$ represents the length of the rectangle, in feet. Therefore, the best interpretation of $15w$ in this context is that it's the length of the rectangle, in feet.

Choice B is incorrect. This is the best interpretation of y, not $15w$, in the given function.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect. This is the best interpretation of w, not $15w$, in the given function.

Q7

D

D. -8

Choice D is correct. By the zero-product property, if $(x-4)(x+2)(x-1)=0$, then $x-4=0$, $x+2=0$, or $x-1=0$. Solving each of these equations for x yields $x=4$, $x=-2$, or $x=1$. The product of these solutions is $(4)(-2)(1)=-8$.

Choice A is incorrect and may result from sign errors made when solving the given equation. Choice B is incorrect and may result from finding the sum, not the product, of the solutions. Choice C is incorrect and may result from finding the sum, not the product, of the solutions in addition to making sign errors when solving the given equation.

Q8

A

A. $\frac{h^{10}}{q^{14}}$

Choice A is correct. For positive values of a, $\frac{a^m}{a^n} = a^{m-n}$, where m and n are integers. Since it's given that $h > 0$ and $q > 0$, this property can be applied to rewrite the given expression as $h^{15-5}q^{7-21}$, which is equivalent to $h^{10}q^{-14}$. For positive values of a, $a^{-n} = \frac{1}{a^n}$. This property can be applied to rewrite the expression $h^{10}q^{-14}$ as $h^{10}\frac{1}{q^{14}}$, which is equivalent to $\frac{h^{10}}{q^{14}}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q9 C

C. 7

Choice C is correct. Substituting -1 for x in the given function f gives $f(-1)=(-1)^3+3(-1)^2-6(-1)-1$, which simplifies to $f(-1)=-1+3(1)-6(-1)-1$. This further simplifies to $f(-1)=-1+3+6-1$, or $f(-1)=7$.

Choice A is incorrect and may result from correctly substituting -1 for x in the function but incorrectly simplifying the resulting expression to $f(-1)=-1-3-6-1$, or -11. Choice B is incorrect and may result from arithmetic errors. Choice D is incorrect and may result from correctly substituting -1 for x in the function but incorrectly simplifying the expression to $f(-1)=1+3+6+1$, or 11.

Q10 B

B. $n = \frac{7m}{5} - p$

Choice B is correct. It's given that the equation $7m = 5n + p$ relates the positive numbers m, n, and p. Dividing both sides of the given equation by 5 yields $\frac{7m}{5} = n + p$. Subtracting p from both sides of this equation yields $\frac{7m}{5} - p = n$, or $n = \frac{7m}{5} - p$. It follows that the equation $n = \frac{7m}{5} - p$ correctly gives n in terms of m and p.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. `0.25` ≡ `1/4`). For grid-in items, decimal and fraction answers are both valid.

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